HOMOLOGY FROM UNIVERSAL PROPERTIES: RESULTS AND A COHERENCE QUESTION

Abstract. We describe a checked development of the beginnings of homology in ordinary dependent type theory with univalence and an axiomatic ambient category of higher categories. Copowers provide the basic construction; a representable stability assumption gives abelian groups, Mayer–Vietoris exactness on pointed components and a suspension equivalence. Excision and the subsequent results currently assume the pentagon for the chosen internal composition. We explain the outstanding problem of deriving this pentagon. A finite recursive construction of exact sequences of groups from pointed fibres is now checked; its homology instance gives the positive-degree Mayer–Vietoris sequence of abelian groups, exact at all three repeating positions. Its degree-zero extension and categorical naturality remain open. We ask whether universal properties can supply a finite, reusable treatment of the coherence required by structural arguments in homology.

1. Purpose and mathematical setting

The aim is a formal language in which arguments using stable higher categories and universal properties retain their usual mathematical form. Homology is a test case: form the colimit of a constant diagram, use stability, and extract homotopy groups. The following is a mathematical account of the results already checked and of the remaining question. Every proposition below is conditional on the hypotheses specified; no construction of a model of all these hypotheses is claimed.

The foundations are ordinary dependent type theory with universes, identity types and univalence. Function extensionality follows from univalence. Image and replacement assumptions provide the propositional truncations and set quotients used below. We write $\pi_0 Y = \|Y\|_0$. Pushouts of types, when needed, are specified by their mapping universal property; no new type-theoretic rules are introduced.

Following Sattler and Wärn's axiomatic approach [3], we are given $\mathsf{Cat} : \mathcal{U}_{\ell+1}$, types $\mathrm{Map}(A, B) : \mathcal{U}_\ell$, identities, composition, and associativity and unit paths. In particular,

$$a_{F,G,K} : (F \circ G) \circ K = F \circ (G \circ K).$$

The types $\mathrm{Map}(A, B)$ need not be sets. We assume specified pentagon and triangle identities for these paths. These are equalities between paths, not strict equations for composition.

The structural axioms give a terminal category $\mathbf{1}$, pullbacks, exponentials, and an interval $\mathbb{I}$ with two objects $0, 1$. The interval classifies monotone maps to the two-element order. The ambient category is univalent; the internal categories satisfy a Rezk univalence axiom; and the interval detects equivalences of categories. There is a Segal axiom in dimension two. The simplices are constructed from the interval, with $\Delta^0 = \mathbf{1}$ and $\Delta^{n+1} = \mathrm{Fun}(\Delta^n, \mathbb{I})$.

For $C : \mathsf{Cat}$, set

$$\mathrm{Ob}\, C = \mathrm{Map}(\mathbf{1}, C), \qquad \mathrm{Hom}_C(x, y) = \sum_{m : \mathrm{Map}(\mathbb{I}, C)} (m \circ 0 = x) \times (m \circ 1 = y).$$

We distinguish ambient composition $F \circ G$ from internal composition: for $f : x \to y$ and $g : y \to z$ write $f \star g : x \to z$, in diagrammatic order. A square in C includes a specified path between its two composites.

Proposition 1.1 (Segal and internal composition)**.** *For every $n \geq 0$, restriction to the spine is an equivalence*

$$\mathrm{Map}(\Delta^n, C) \simeq \sum_{x_0, \ldots, x_n : \mathrm{Ob}\, C} \prod_{i=0}^{n-1} \mathrm{Hom}_C(x_i, x_{i+1}).$$

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Consequently, the type of simplex fillers of each prescribed spine is contractible. Internal composition, unit laws and a specified associator

$$\alpha_{f,g,h} : (f \star g) \star h = f \star (g \star h)$$

have been constructed. Functors preserve identities and composition up to specified paths. With the univalence and detection axioms, fully faithful functors reflect invertibility and induce embeddings on object types; fully faithful, essentially surjective functors are equivalences.

These results use the ambient pentagon and triangle, but no ambient coherence above them. In particular, the quantifier over n in the Segal theorem does not require successively higher coherence assumptions. The pentagon for the *chosen internal associator* has not yet been derived.

2. Copowers and excision

Fix $D : \mathsf{Cat}$. Let $\mathcal{U}_k$ be a universe strictly smaller than $\mathcal{U}_\ell$. For $X : \mathcal{U}_k$ and $A : \operatorname{Ob} D$, assume an object $X \otimes A$ and arrows $\eta_x : A \to X \otimes A$, such that, for every $T : \operatorname{Ob} D$,

$$r_{X,A,T} : \operatorname{Hom}_D(X \otimes A, T) \xrightarrow{\simeq} (X \to \operatorname{Hom}_D(A, T)), \qquad g \longmapsto (x \longmapsto \eta_x \star g). \tag{1}$$

This is the mapping universal property of the colimit of the constant X-diagram at A. The size restriction is part of the assumptions.

Proposition 2.1 (Basic copower properties)**.** *A map $f : X \to Y$ induces $t_A(f) : X \otimes A \to Y \otimes A$, with $\eta_x \star t_A(f) = \eta_{f(x)}$. There are paths*

$$t_A(\mathrm{id}) = \mathrm{id}, \qquad t_A(g \circ f) = t_A(f) \star t_A(g).$$

Homotopic maps have equal images, and equivalences induce invertible arrows. Moreover,

$$\mathbf{1} \otimes A \simeq A, \qquad \varnothing \otimes A \text{ is initial},$$

$$(X + Y) \otimes A \simeq (X \otimes A) \amalg (Y \otimes A), \qquad (X \times Y) \otimes A \simeq X \otimes (Y \otimes A).$$

Here $\simeq$ between objects denotes equivalence in D.

Argument. Invert (1) to construct the arrows. Their comparisons follow by restricting along every η_x, using the inverse's computation path, associativity and function extensionality. The coproduct and Fubini assertions apply the same argument to families indexed by $X + Y$ and $X \times Y$. □

The coproduct statement is binary. These comparisons have been proved individually; they are not an assertion that an entire coherent copower functor has already been constructed.

Hypothesis 2.2 (Internal pentagon P_D)**.** For four composable arrows in D, the two usual composites of the specified associator α from $((f \star g) \star h) \star k$ to $f \star (g \star (h \star k))$ are equal:

$$\alpha_{f\star g,h,k} \cdot \alpha_{f,g,h\star k} = \big((\alpha_{f,g,h} \star k) \cdot \alpha_{f,g\star h,k}\big) \cdot (f \star \alpha_{g,h,k}).$$

Whiskering of paths is denoted by the same symbol $\star$.

Proposition 2.3 (Excision, assuming P_D)**.** *A pushout square of small types*

$$\begin{array}{ccc} C & \xrightarrow{\ i\ } & B \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle u} \\ E & \xrightarrow[\ v\ ]{} & X \end{array}$$

is carried by t_A to a pushout square in D, with the specified commutativity path induced by the square and the copower comparisons. The pushout of types is required to have its universal property for targets in $\mathcal{U}_\ell$.

Argument. For $T : \operatorname{Ob} D$, compare cocones from the image square into T with cocones from the original square into $\operatorname{Hom}_D(A, T)$. Restriction (1) gives equivalences on the two legs and on their path data. The resulting comparison square commutes. The verification on the path data uses P_D: it compares reassociations of four internal arrows, including a generator η_c. The pushout universal property for types then gives the required equivalence for D. □

3. Groups, exactness and suspension

For this section assume P_D and the following additional data: a zero object 0_D; a chosen suspension ΣB, given by a pushout of $0_D \leftarrow B \rightarrow 0_D$; and an object $\mathbb{S} \in D$. No monoidal structure or generator property of $\mathbb{S}$ is assumed.

The stability assumption is specifically the following representable condition: for every pushout square in D and every $S \in D$, applying $\mathrm{Hom}_D(S, -)$ gives a pullback of types. We do not assume that all finite colimits in D exist. The results concern the pushouts provided above or given in the hypotheses.

Proposition 3.1 (Loop equivalences)**.** *There are equivalences*

$$L_{B,T} : \mathrm{Hom}_D(\Sigma B, T) \simeq \Omega(\mathrm{Hom}_D(B, T), 0),$$
$$\sigma_{B,T} : \mathrm{Hom}_D(B, T) \simeq \Omega(\mathrm{Hom}_D(B, \Sigma T), 0).$$

The first follows from the suspension universal property; the second uses stability. The first carries the zero arrow to the constant loop and is natural under postcomposition, with the basepoint change supplied by $0 \star f = 0$.

The naturality statement for L uses P_D. For $n \geq 0$, define

$$H_n(X; A) = \pi_0 \,\mathrm{Hom}_D(\Sigma^n \mathbb{S}, X \otimes A).$$

Proposition 3.2 (Groups and positive-degree homomorphisms)**.** *Each $H_n(X; A)$ has a specified abelian group structure. For $n \geq 1$, the maps $H_n(f; A)$ induced by $t_A(f)$ are homomorphisms for these structures. Identity and composition laws for the underlying maps of sets hold in every degree.*

Argument. In positive degrees transport the fundamental group structure along $\pi_0 L$. In degree zero use $\pi_0 \sigma$ instead. Stability supplies the further deloopings needed for the Eckmann–Hilton argument, proving commutativity in every degree. For positive-degree homomorphisms, naturality of L identifies postcomposition with the loop map of a pointed map. Such a map preserves loop multiplication. Transport this fact along $\pi_0 L$. □

This last argument needs no coherence relating the chosen naturality paths of L for two successive maps. The homomorphism assertion in degree zero, and the comparison of the two available group structures on positive-degree groups, remain unproved.

Mayer–Vietoris exactness on components. For the square in Proposition 2.3, a connecting map

$$\partial : H_{n+1}(X; A) \longrightarrow H_n(C; A)$$

has been constructed. The following statements are proved on sets of components, pointed by the zero arrows:

(1) $u_* i_* = v_* j_*$. If $u_* b = v_* e$, then there merely exists c with $i_* c = b$ and $j_* c = e$.
(2) $i_* \partial = 0$ and $j_* \partial = 0$. If $i_* c = 0 = j_* c$, then there merely exists z with $\partial z = c$.
(3) $\partial(u_* b - v_* e) = 0$ for $b \in H_{n+1}(B; A)$ and $e \in H_{n+1}(E; A)$. Conversely, if $\partial z = 0$, there merely exist such b, e with $z = u_* b - v_* e$. This difference uses the specified group structure in degree $n + 1$.

Here "merely exists" is propositional existence. Indeed, stability identifies $\mathrm{Hom}_D(S, C \otimes A)$ with the type of triples $(b, e, p : u_* b = v_* e)$. The connecting map is obtained from triples $(0, 0, p)$, using L to supply p. The boundary preserves the zero class. These results establish the three repeating exactness positions, but do not yet give a natural long exact sequence of abelian groups: additivity of the chosen boundary, the degree-zero homomorphisms and categorical naturality remain open. Boundary naturality has been proved for maps of pointed pullback diagrams of types with specified cospan homotopies; its application to the categorical diagrams still needs the comparison data.

A structural construction, described in the next section, now gives a replacement boundary with additivity and exactness on both sides proved together in positive degrees. Comparison with the original formula is not needed for that result and has not been proved.

Reduced groups and spheres. Write $! : X \to \mathbf{1}$, and define

$$\mathrm{Red}(S, X; A) = \mathrm{fib}_0\big(\mathrm{Hom}_D(S, X \otimes A) \xrightarrow{t_A(!)_*} \mathrm{Hom}_D(S, \mathbf{1} \otimes A)\big),$$
$$\widetilde{H}_n(X; A) = \pi_0 \,\mathrm{Red}(\Sigma^n \mathbb{S}, X; A).$$

For a suspension SX specified by the pushout $\mathbf{1} \leftarrow X \to \mathbf{1}$, the equivalence

$$\mathrm{Red}(\Sigma B, SX; A) \simeq \mathrm{Red}(B, X; A)$$

has been proved. Hence $\widetilde{H}_{n+1}(SX; A) \simeq \widetilde{H}_n(X; A)$ as sets. With chosen desuspensions in D, satisfying $\Sigma(\Sigma^{-1}B) \simeq B$, the resulting sphere computations include

$$\widetilde{H}_{n+j}(S^n; A) \simeq \pi_0 \,\mathrm{Hom}_D(\Sigma^j \mathbb{S}, A) \quad (n, j \geq 0),$$

and

$$\widetilde{H}_k(S^{k+m}; A) \simeq \pi_0 \,\mathrm{Hom}_D(\Sigma^{-m} \mathbb{S}, A) \quad (k, m \geq 0).$$

Here $S^0 = S\varnothing$ and $S^{n+1} = SS^n$. These are bijections of sets; compatibility with group structures is not yet proved.

The intended example is spectra, with $\mathbb{S}$ the sphere spectrum and $X \otimes A \simeq \Sigma^\infty X_+ \wedge A$ [2]. Taking $A = H\mathbb{Z}$ should recover ordinary integral homology. Neither this model nor this comparison has been constructed in the development. In particular, $\mathbf{1} \otimes A \simeq A$ is a coefficient normalization, not the dimension axiom asserting the vanishing of positive homology of a point. Such vanishing requires further information about A.

4. One fibre construction supplies an exact sequence

The following structural result has been checked using only the type-theoretic foundations. Start with one pointed map $f_0 : (Y_0, y_0) \to (Y_{-1}, y_{-1})$. Recursively define

$$Y_{r+1} = \mathrm{fib}_{y_{r-1}}(f_r), \qquad f_{r+1} : Y_{r+1} \longrightarrow Y_r$$

by the fibre projection, with its canonical basepoint.

Proposition 4.1 (Exact fibre tower)**.** *For every $r \geq 0$, the sequence*

$$\pi_1 Y_{r+1} \xrightarrow{(f_{r+1})_*} \pi_1 Y_r \xrightarrow{(f_r)_*} \pi_1 Y_{r-1}$$

consists of group homomorphisms and is exact at $\pi_1 Y_r$. A specified commuting square of pointed initial maps, including its basepoint compatibility, induces commuting squares of groups at every stage.

Argument. The equivalence $\Omega\,\mathrm{fib}(f) \simeq \mathrm{fib}(\Omega f)$ is proved together with its compatibility with the projection to ΩY. It therefore gives both the null-composite and the lifting assertion for the same loop maps. Effectiveness of set truncation gives exactness on components; these maps are homomorphisms because they are loop maps. Iteration is ordinary recursion on $\mathbb{N}$. A pointed square induces a square of fibre inclusions. Its new basepoint compatibility follows from the first-projection computation of the path of pairs; it is not additional input. Applying this construction recursively proves the last assertion. □

The second fibre is identified with ΩY_{-1}, including its map into the first fibre. More generally, for a pointed homotopy pullback $Q \simeq B \times_P E$, projection $Q \to B \times E$ has fibre equivalent to ΩP, over the boundary map into Q. This gives one connecting homomorphism $\pi_2 P \to \pi_1 Q$ whose image is the kernel of the product projection. The map and its exactness come from the same fibre inclusion.

Apply this to the representable pullback supplied by stability for the excision square, with probe $\Sigma^n \mathbb{S}$. The pointed suspension–loop equivalences identify the source and target groups with the existing homology groups. The projection paths identify the outgoing maps with the existing i_* and j_*. The following stronger statement is now checked.

Proposition 4.2 (Positive-degree Mayer–Vietoris)**.** *There are homomorphisms $\partial^{\mathrm{fib}} : H_r(X;A) \to H_{r-1}(C;A)$ for every $r \geq 2$. For every such r, the sequence*

$$H_r(B;A) \times H_r(E;A) \xrightarrow{u_*-v_*} H_r(X;A) \xrightarrow{\partial^{\mathrm{fib}}} H_{r-1}(C;A) \xrightarrow{(i_*,j_*)} H_{r-1}(B;A) \times H_{r-1}(E;A) \xrightarrow{u_*-v_*} H_{r-1}(X;A)$$

consists of homomorphisms of the specified abelian groups and is exact at all three interior terms. The arrows agree as r varies, giving the positive-degree exact sequence ending at $H_1(X;A)$; exactness at this last term is not asserted.

Argument. For a family $V : Y \to \mathcal{U}$, inclusion of a fibre $V(y) \to \sum_{x:Y} V(x)$ has fibre over (y, v) equivalent to ΩY: the equivalence sends a path of pairs to its projection in Y. Applying this to the pullback cone identifies the fibre of $\Omega P \to Q$ with $\Omega(B \times E)$. Its map to ΩP is the difference of the two loop maps. Hence the same fibre-sequence theorem gives exactness at $\pi_2 P$, and looping the difference gives the usual group difference on π_2.

One general lemma simplifies the compatibility argument. If loops at $z : Z$ commute, conjugation by any loop at z is the identity. Thus transport of loops along two paths $x = z$ agrees. Homotopic maps into (Z, z) therefore induce equal loop maps with arbitrary choices of pointing paths; no compatibility between those pointing paths and the homotopy is required. Eckmann–Hilton applies this to loop spaces. Together with the already proved naturality of L, it identifies the maps on π_2 with the usual homology maps, without an additional coherence law for L.

Exactness at $H_{r-1}(C;A)$ is the fibre inclusion theorem above. Exactness at the product follows directly from the representable pullback: $u_*b - v_*e = 0$ is equivalent to $u_*b = v_*e$. Finally, difference of homomorphisms into an abelian group is a homomorphism. □

No coherence assumption beyond P_D has been introduced. These results establish the conventional positive-degree sequence with one boundary, without comparing it to the earlier direct formula. Extending this sequence through degree zero, and establishing its naturality for the intended maps of categorical or copower diagrams, remain open. The recursive fibre tower itself already has naturality for specified pointed input squares.

5. Coherent uniqueness of universal witnesses

There is a general, fully proved coherence result in ordinary type theory. If $e : V \simeq W$ and $w : W$, the type

$$\sum_{v:V} (e(v) = w)$$

is contractible. This includes the computation path as part of the witness. Every iterated identity type with a specified globular boundary in this fibre is also contractible.

This assertion has a finite formulation. For a type Z, define boundary types B_n and cell families $E_n : B_n \to \mathcal{U}$ recursively by

$$B_0 = \mathbf{1}, \quad E_0(*) = Z, \qquad B_{n+1} = \sum_{b:B_n} E_n(b) \times E_n(b), \quad E_{n+1}(b,p,q) = (p = q).$$

Contractibility of Z implies

$$\prod_{n:\mathbb{N}} \prod_{b:B_n} \mathrm{isContr}(E_n(b)).$$

The proof is induction, using that identity types in a contractible type are contractible. Dependent families of contractible choices likewise have a contractible type of sections.

For copowers this applies to

$$W(h) = \sum_{g:\mathrm{Hom}_D(X \otimes A, T)} (r_{X,A,T}(g) = h).$$

Identity and composition comparisons of universal witnesses have been constructed in such fibres, retaining the computation paths. These comparisons have not all been identified with the previously chosen comparison paths after forgetting the witnesses.

Transport of witnesses along specified commuting squares is also proved. Given $r : V \to W$, $s : V' \to W'$, maps $u : V \to V'$, $v : W \to W'$, and $\theta_x : s(u(x)) = v(r(x))$, transport is

$$(x, p) \longmapsto (u(x), \theta_x \cdot \mathrm{ap}_v(p)) : \mathrm{fib}_w(r) \longrightarrow \mathrm{fib}_{v(w)}(s).$$

Identity and pasting laws follow from identity elimination. If s is an equivalence, the type of maps between these fixed fibres is contractible, so their higher comparisons are available as well. The outstanding issue is to supply the categorical boundary data needed to apply this principle, compatibly with the specified operations.

6. The internal pentagon: a precise outstanding step

The contractible type of four-simplex fillers already exists. The difficulty is identifying the chosen associators with the comparisons read from its faces. For a simplex $T : \Delta^4 \to C$ and face maps $d : \Delta^3 \to \Delta^4$, $d' : \Delta^2 \to \Delta^3$, reading an iterated face involves comparing

$$(T \circ d) \circ d' \quad \text{and} \quad T \circ (d \circ d').$$

These comparisons use the ambient associator, and compatibility among them involves its higher coherence.

For clarity, call the ambient associativity and unit paths level 1, and their pentagon and triangle level 2. A specified level 3 axiom K_5 has been formulated: for five composable functors, the two pastings of pentagon faces and associator naturality faces of the three-dimensional associahedron K_5 are equal. This is an equality between two equalities of paths.

Question 6.1 (First coherence test)**.** *Do the stated ambient axioms through level 2, together with* K_5*, imply the pentagon for the particular internal associator constructed from Segal fillers, for every* $C : \mathsf{Cat}$*?*

A proposed proof first compares the action of three successive functors on an internal morphism. For $F : C \to D$, $G : B \to C$, $K : A \to B$ and $m : \mathbb{I} \to A$, the underlying morphism paths agree by the ambient pentagon at (F, G, K, m). Their endpoint comparisons are expected to agree by K_5 at $(F, G, K, m, 0)$ and $(F, G, K, m, 1)$. Both actual paths have now been expressed as paths in Hom over the same underlying morphism path, leaving two explicit endpoint equations. Their identification with the K_5 pastings remains open.

One must then pass to triangle boundaries and composition witnesses, prove preservation of the chosen associator under functors, and apply this to a four-simplex. Reduction of the internal pentagon to the four-simplex spines has been proved; its value on those spines has not. Thus Question 6.1 is a proposed theorem, not a result already available. All results above invoking P_D still assume it independently.

Univalence does not by itself settle this point. It permits transport along equivalences, whereas the face maps are generally not equivalences. A universal characterization might allow coherence to be transported, but the compatibility of the independently specified ambient composition has to be justified.

7. The open question for homology

The broader objective is suggested by synthetic category theory [1]: check proofs at the level of universal properties, stability and descent. For homology the desired argument is

copowers preserve pushouts

$\Downarrow$ stability

pullbacks of mapping spaces

$\Downarrow$ homotopy groups

exact sequences of homology groups.

Each implication must include the relevant functoriality and compatibility of the specified comparisons.

Question 7.1 (A finite structural treatment of homology)**.** *Can one give, within ordinary dependent type theory with univalence, a finite collection of structural data and universal properties from which the coherence needed for this argument follows uniformly? More specifically, can such a presentation yield the copower construction, its excision and suspension equivalences, and natural long exact sequences of abelian groups, with compatibility of the universal comparisons proved once rather than assumed separately as new higher-dimensional laws?*

"Finite" here allows quantification over natural numbers and finite diagrams, as in the contractibility theorem of Section 5. It asks for actual expressions and proofs in the existing type theory, without judgmentally associative composition for Cat or new directed type formers. Merely postulating that all the needed operations are coherent would leave the problem unchanged.

There are two possible scales of answer. For homology alone, a fixed finite amount of ambient coherence may suffice. Question 6.1 is the first unresolved test. Section 4 now supplies a finite recursive construction of exact sequences of groups, and its positive-degree homology instance uses only the existing P_D. There is presently no established bound for the remaining homology statements. A stronger answer would characterize ambient composition, or at least its action on simplex boundaries, by universal data whose compatibility types are contractible. This would explain systematically how coherent uniqueness of fillers becomes coherence of the categorical constructions extracted from them.

Such a result should also admit an interpretation in the intended higher-categorical models. For ordinary homology, one would additionally need a suitable coefficient object and a comparison or normalization theorem. More broadly, the same structural language should eventually support arguments about stable categories and descent for families of higher categories or stacks. Those extensions are motivations, not results of the present development. The immediate question is whether homology can supply a first substantial example in which the coherence cost is absorbed into reusable mathematical principles.

References

[1] D.-C. Cisinski, B. Cnossen, K. Nguyen and T. Walde, *Synthetic Category Theory*, book project, version of 12 July 2026. Available online.

[2] J. Lurie, *Higher Algebra*, version of 18 September 2017, Chapter 1, especially §§1.1 and 1.4. Available online.

[3] C. Sattler and D. Wärn, axioms for higher category theory, as presented in Wärn's notes *Higher-categorical headcanon*. Notes and slides linked from the author's webpage.